

1.0 Introduction

The Studs (hollow studs) of base oil filters and fuel in 3500 Series engines are installed with difficulty using pipe wrenches, not applying the required torque; running the risk of being lazy and in the worst case damage the threads of these, causing rework and wasted time (see Figure 1a, 1b, 1c.)



Figure 1.a. Missing Stud in base oil filter.



Figure 1.b. The stud came off with the filter after improper installation.



Figure 1.c. Stud installation with pipe wrench, with this tool is most likely to damage the threads.

This Best Practice aims to facilitate the development of this task to design a user-friendly, safe and quick use.

2.0 Best Practice Description

The required torque without the risk of damaging the threads: a toolkit for proper installation of the studs of the base oil filters and fuel, which allows installing properly designed.

Several tools were manufactured according to the length and diameter of each stud. The tool consists of two main parts, the top and both steel socket adapter ASTM4140.

The cover surrounds the socket adapter to join the two symmetrical parts of this eventually forming the hexagonal head 9/16 overhanging a superior hole in the lid and where the torque meter is installed. It is bringing threaded stud inside it will be installed.

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On the other hand the lid is fitted with two forcing screws on each side of the socket adapter, once these allow the stud installed easily remove the cover. (See Figure 2.)

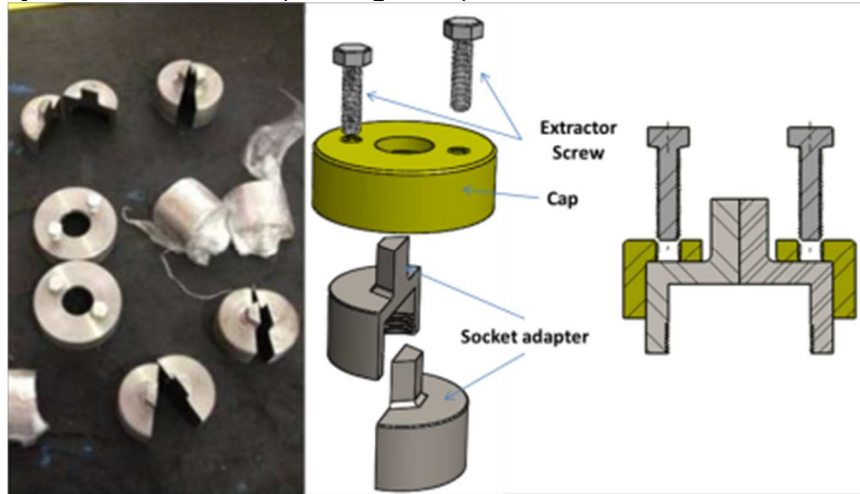


Figure 2. Architecture Tool.

3.0 Implementation Steps

Then the use of the tool is detailed.

The tool is assembled by joining the two halves of the socket adapter and placing the lid on top. The screws are tightened only to touch the upper surface of the socket adapter.



Figure 3.

The stud is screwed into the socket adapter to take him to stop.



Figure 4.

Loctite 271 is applied at the end of the thread, as indicated in the SIS manual and the stud is installed in the base. Finally torque required by this applies.

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Specification for 151-9270 Oil Filter Base As and 8N-2001 Oil Filter Base As			
Item	Qty	Part	Specification Description
A	-	-	Apply Loctite 271 on the tapered end of the hollow stud to a distance of 10.2 ± 1.5 mm (0.40 ± 0.06 inch).
1	1	2P-4304 Hollow Stud	Torque to 80 ± 14 N·m (59 ± 10 lb ft).

Figure 5. Example of installation of filter base stud (2P-4304) using tooling as per specifications within SENR1122.

The cap is removed by inserting the tool extractor screws, once a loose socket adapter halves are separated and thus the tool is fully removed and installed with the exact torque stud.



4.0 Benefits

- Easy fabrication
- Easy operation
- Prevents damage to the threads of the stud
- Lightweight for easy transport (with maximum weight of 0.5kg)

5.0 Resources Required

Plans and sheet materials of the tool (see attached map)
Approximate cost of 110USD tool. c / u

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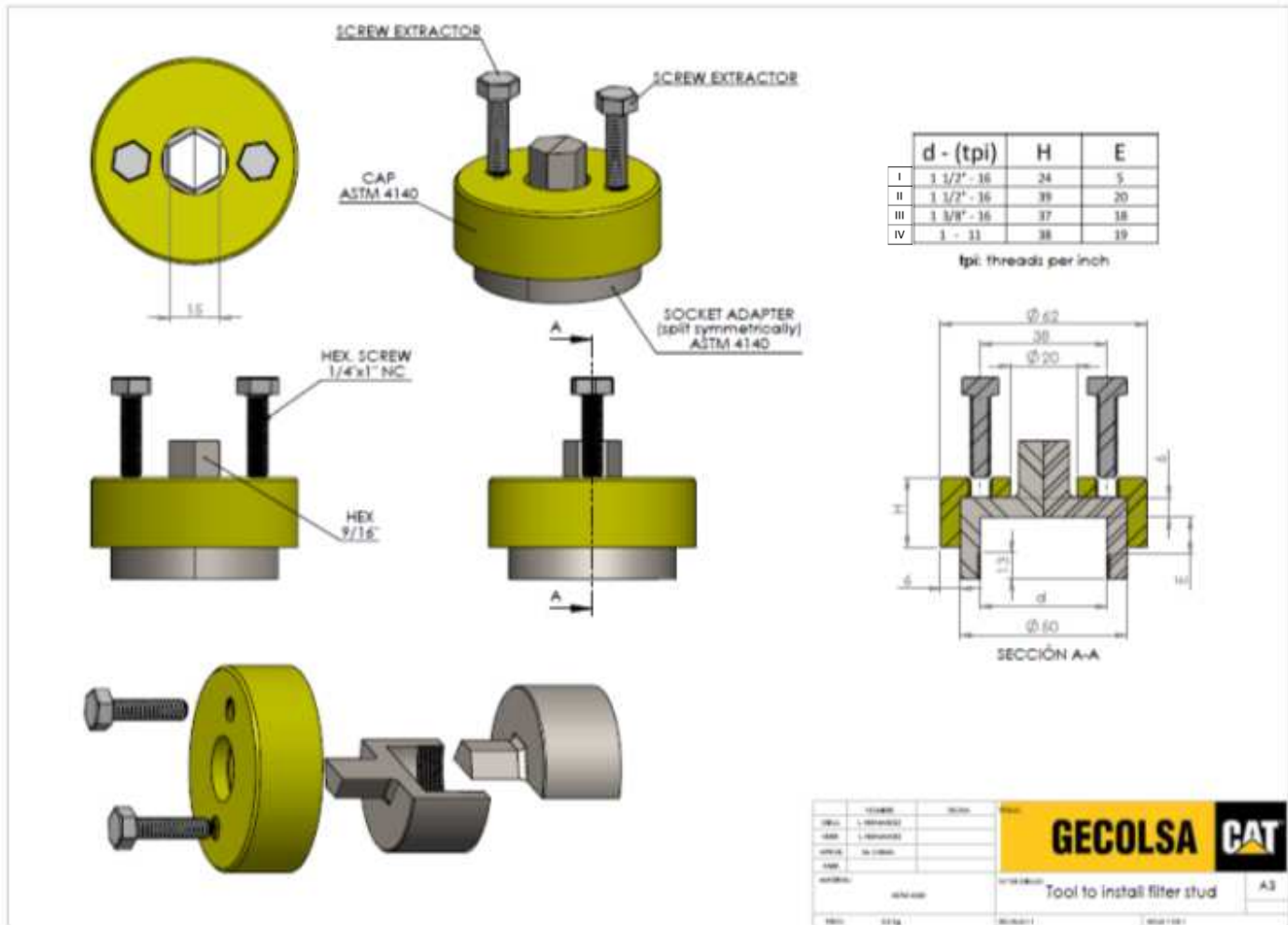
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6.0 Supporting Attachments / References



Tool	Stud P/N	Application	Torque (lb ft)	Media Number
I	2W-8225	Oil	60+-10	SENR9936
II	2P-4304	Oil	60+-10	SENR1122
III	8N-3653	Fuel	52+- 11	KENR9885
IV	9S-4167	Fuel	52+- 11	KENR9885

The yellow cap and the extractor screws are common for all the tools. The socket adapters are different for each tool and they depend on the studs dimensions. The dimensions for each socket adapter are specified in the drawing.

7.0 Related Best Practices

None

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8.0 Acknowledgements

Enrique Urbina
CRC Development Projects Chief, Relianz
Email: enrique.urbina@relianz.com.co

Wilson Arias
Supervisor Técnico, Relianz.
Email: Wilson.arias@relianz.com.co

Checked and written by:

David Antonio Gonzalez Fernandez
Ingeniero de Servicio CRC, Relianz
Email: antonio.gonzalez@relianz.com.co

Melissa Chimá B.
Ingeniera de Servicio CRC, Relianz.
Email: melissa.chima@relianz.com.co

Luis Alberto Fernández Ch.
Ingeniero de Desarrollo
Relianz CAT División Minería
Departamento Técnico, área Ingeniería y Desarrollo.
Email: luisalberto.fernandez@relianz.com.co

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